

AMANI PENUMARTHI

Bangalore, India (Open to Relocate to Germany)

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Summary

Aerospace Electrical Design Engineer with 5+ years across Boeing 747, 737 and 777 programs, specialising in EWIS and wire harness design. Experienced in routing bundles in CATIA V5, running DMU and node checks and managing configurations in ENOVIA and 3DEXperience. Converts legacy BCA and 2D inputs into manufacturable harness models, foamboards and WHIMST BOMs with 100% on time and 97 to 99% quality delivery. Applies EWIS separation rules, backshell clocking and grounding practices to maintain safe, compliant installations for cabin interiors and systems. Known for checklist-driven quality control, analytical problem solving and clear collaboration with design, stress, installation and manufacturing teams.

Core Skills

- **Electrical Design:** EWIS validation, WHI & WHA architecture, routing, separation analysis, backshell clocking, grounding
- **CAD & PLM:** CATIA V5, CATIA V4, Siemens NX, ENOVIA PLM, 3DEXperience, DMU analysis, configuration & change control
- **Aircraft Domain:** Boeing 747, 737, 777 cabin interiors, EWIS rules, installation notes, harness manufacturing preparation
- **Quality & Methods:** Design rule checks, EMA reports, WHIMST BOM, GBN, MBS, WHP data, DFM, DFMEA, defect prevention
- **Delivery & Process:** Product lifecycle management, on time delivery, high first time right quality, process optimisation
- **Collaboration:** Cross-functional teamwork, documentation, mentoring support, troubleshooting, continuous improvement mindset

Experience

Cyient Ltd

August 2023 – Present

Design Engineer

Bangalore, India

- Own EWIS and harness design for Boeing 747 and 777-9 cabin interiors and systems from concept to release.
- Model and route wire bundles in CATIA V5, run DMU and node checks, maintain configuration in ENOVIA PLM.
- Prevent routing clashes and missing terminations and cut late harness defects by around 30% across 50+ segments.
- Deliver 20+ flattening packages and WHIMST BOMs with 100% on time release and 97%+ internal quality scores.

San Engineering Solutions

April 2023 – August 2023

Design Engineer

Bangalore, India

- Owned part creation and harness integration for Boeing programs to improve reuse, quality and harness build readiness.
- Converted V4 and PI inputs into CATIA V5 WHI bundles, applying connectors, clamps, backshells and routing rules.
- Built GBN, MBS and WHP data with validated attributes to support network validation, installation planning and review.
- Resolved 95% of EMA rule issues before review and kept PLM models clean, enabling 100% on time and 99%+ quality.

K-Tech Centre of Excellence in Aerospace and Defence, KSCST

April 2022 – April 2023

Junior Design Engineer

Bangalore, India

- Supported WHI harness conversion for Boeing 737, rebuilding installations in CATIA V5 from BCA legacy sources.
- Applied installation rules and separation codes while routing bundles and supports to keep clearances to structures.
- Prepared manufacturing-ready harness models and drawings so teams could build and install with fewer late changes.
- Standardised conversion steps and checklists with senior engineers, cutting harness conversion time by 10-15%.

- Taught mechanics of fluids and aerodynamics to undergraduate mechanical and aeronautical engineering students.
- Prepared lectures, problem sets and lab support that reinforced understanding of core fluid and aero concepts.
- Linked theoretical topics with practical aerospace design decisions used later in modelling, simulation and analysis.

Central Manufacturing Technology Institute**2019**

- Worked with the hydraulic systems team to study flow behaviour and component performance in industrial applications.
- Designed flow divider and combiner valve concepts in CAD and evaluated feasibility with senior engineering guidance.
- Documented methods, calculations and design outcomes in a technical report adopted as reference by the host team.

Education

Visvesvaraya Institute of Advanced Technology, VTU Muddenahalli**2018 – 2020****Acharya Institute of Technology, VTU****2014 – 2018****Languages**

- English – Full Professional Proficiency (C2)
- German – Intermediate Working Proficiency (B1)